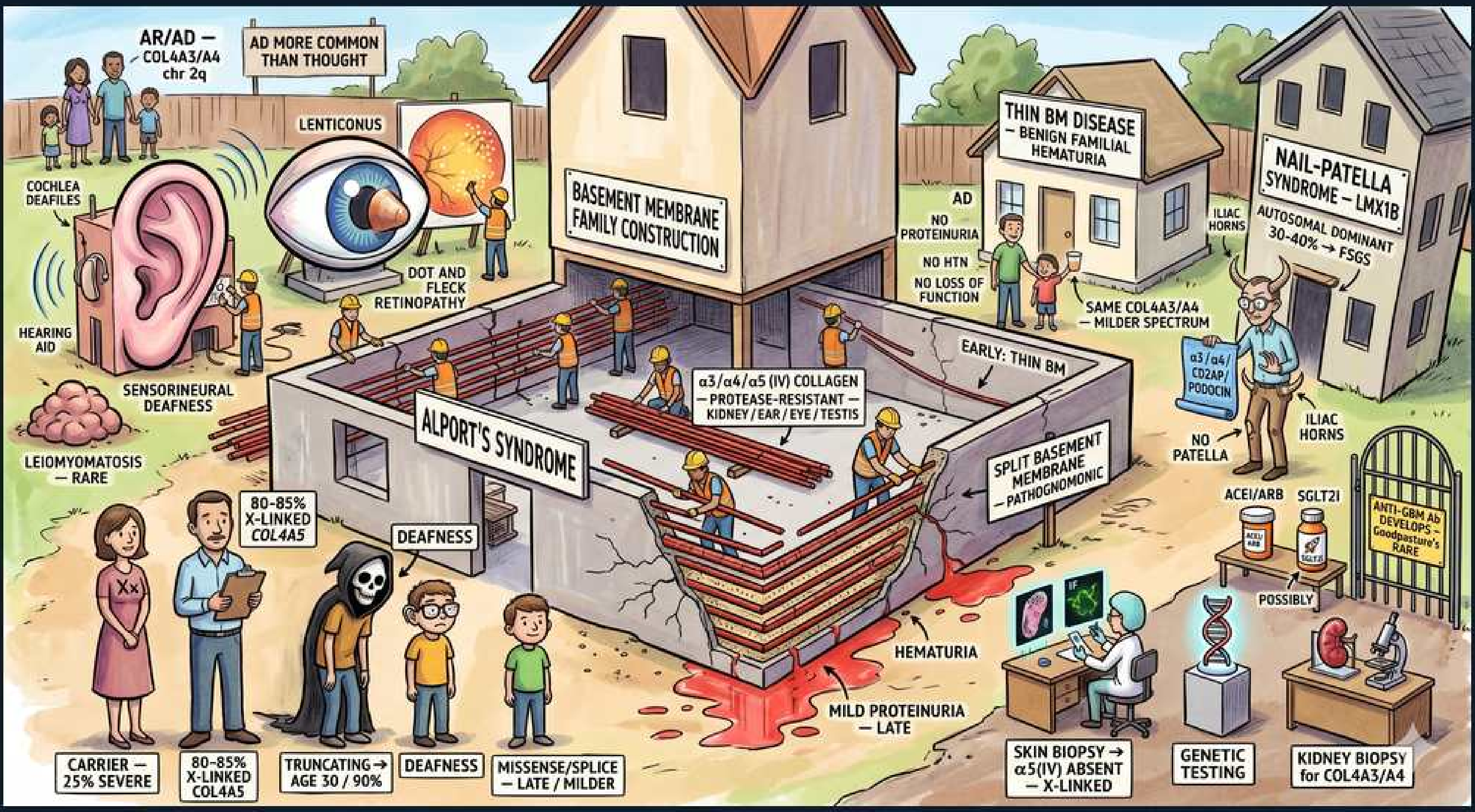
Basement Membrane Syndromes — Alport, Thin BM & Nail-Patella



1. Alport's syndrome — type IV collagen

WHAT YOU SEE

Banner 'Alport's syndrome', α3/α4/α5 (IV) collagen

WHAT IT ENCODES

Alport syndrome is a hereditary defect in type IV collagen (α3/α4/α5 chains, COL4A3/4/5) that disrupts the GBM, cochlea, and lens basement membranes. It causes progressive hematuric nephritis, sensorineural deafness, and ocular changes.

BOARD TRAP

Alport is a structural type IV COLLAGEN defect (GBM), not an immune-complex disease — there are no immune deposits and complement is normal.

2. X-linked COL4A5 (80-85%)

WHAT YOU SEE

'80-85% X-linked COL4A5'

WHAT IT ENCODES

The most common form (~80–85%) is X-linked from COL4A5 mutations. Affected males have severe progressive disease (ESKD, deafness); carrier females are often milder but ~25% can be significantly affected. Autosomal recessive (COL4A3/4) forms also exist.

BOARD TRAP

X-linked Alport hits males hardest; female carriers are usually milder — but a meaningful minority of carriers still progress, so they are not entirely spared.

3. Carrier 25% severe

WHAT YOU SEE

Female figures 'carrier 25% severe'

WHAT IT ENCODES

In X-linked Alport, female carriers have variable severity due to X-inactivation; roughly a quarter develop significant proteinuria, hypertension, or ESKD over time. They require monitoring rather than reassurance.

BOARD TRAP

Don't reassure a female X-linked Alport carrier that she's unaffected — about 25% develop significant kidney disease and need follow-up.

4. Hematuria (early, persistent)

WHAT YOU SEE

Red beam 'hematuria'

WHAT IT ENCODES

Persistent microscopic hematuria, often with episodic gross hematuria, is the earliest and most constant finding, typically from childhood. Proteinuria and hypertension develop later as the GBM deteriorates.

BOARD TRAP

Isolated persistent hematuria with a family history of deafness/renal failure suggests Alport — not benign hematuria; pursue family history and genetics.

5. Mild proteinuria — late

WHAT YOU SEE

'mild proteinuria — late'

WHAT IT ENCODES

Proteinuria appears later than hematuria and can become significant/nephrotic-range as disease advances, heralding decline toward ESKD in affected males. The trajectory is progressive in classic (truncating) mutations.

BOARD TRAP

Hematuria precedes proteinuria in Alport — significant proteinuria signals progression, not the initial presentation.

6. Sensorineural deafness

WHAT YOU SEE

Ear + hearing aid 'sensorineural deafness'

WHAT IT ENCODES

Bilateral high-frequency sensorineural hearing loss results from defective type IV collagen in the cochlear basement membrane. It typically develops in late childhood/adolescence and parallels renal severity.

BOARD TRAP

Alport deafness is SENSORINEURAL (cochlear collagen), not conductive — and it is bilateral and progressive, paralleling renal decline.

7. Anterior lenticonus / dot-fleck retinopathy

WHAT YOU SEE

Eye 'lenticonus', 'dot and fleck retinopathy'

WHAT IT ENCODES

Ocular findings include anterior lenticonus (conical lens deformity, nearly pathognomonic) and dot-and-fleck retinopathy. These eye signs help clinch the diagnosis when combined with hematuria and deafness.

BOARD TRAP

Anterior lenticonus is essentially pathognomonic for Alport — its presence with hematuria + deafness confirms the diagnosis without needing biopsy.

8. Split basement membrane (EM)

WHAT YOU SEE

'split basement membrane — pathognomonic', basket-weave

WHAT IT ENCODES

EM shows the pathognomonic 'basket-weave' pattern: diffuse thickening with splitting and lamellation of the GBM. IF for the α5(IV) collagen chain is absent in X-linked disease, a useful diagnostic test on skin or kidney biopsy.

BOARD TRAP

The basket-weave/split GBM on EM is Alport; uniformly THIN GBM without splitting is thin basement membrane disease — opposite ultrastructure.

9. Thin BM disease — benign familial hematuria

WHAT YOU SEE

'Thin BM disease — benign familial hematuria'

WHAT IT ENCODES

Thin basement membrane nephropathy (benign familial hematuria) is a heterozygous COL4A3/4 defect causing diffusely thin GBM. It presents with isolated persistent microscopic hematuria, NO progression, no proteinuria, no deafness, and normal renal function.

BOARD TRAP

Thin BM disease is benign with NO proteinuria/loss of function or deafness — distinguishing it from early Alport prevents needless alarm; it does not progress to ESKD.

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10. Nail-patella syndrome — LMX1B

WHAT YOU SEE

'Nail-patella syndrome — LMX1B, autosomal dominant'

WHAT IT ENCODES

Nail-patella syndrome (LMX1B mutation, autosomal dominant) causes dysplastic nails, absent/hypoplastic patellae, iliac horns, and a nephropathy with fibrillary collagen deposits in the GBM. ~40% have renal involvement that can progress to FSGS/ESKD.

BOARD TRAP

Nail-patella is LMX1B (collagen in GBM with iliac horns), distinct from Alport's type IV collagen defect — look for absent patellae and iliac horns.

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11. Iliac horns (nail-patella)

WHAT YOU SEE

'iliac horns' gate

WHAT IT ENCODES

Bilateral posterior iliac horns (bony projections from the iliac wings) are pathognomonic radiographic findings in nail-patella syndrome, alongside the nail and patellar dysplasia. They confirm the diagnosis on plain film.

BOARD TRAP

Iliac horns are specific to nail-patella syndrome — they are not seen in Alport or thin BM disease.

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12. Protease-resistant / kidney-ear-eye-testis

WHAT YOU SEE

Beam 'protease-resistant, kidney/ear/eye/testis'

WHAT IT ENCODES

The α3/α4/α5(IV) collagen network is normally protease-resistant and is expressed in kidney GBM, cochlea, lens, and testis — explaining the multi-organ phenotype of Alport. Defective collagen makes these membranes fragile.

BOARD TRAP

Alport's multi-organ involvement (ear, eye, kidney) follows the tissue distribution of α3-5 type IV collagen — a single collagen defect, not separate diseases.

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13. ACEi/ARB; SGLT2 — slow progression

WHAT YOU SEE

'ACEi/ARB, SGLT2'

WHAT IT ENCODES

There is no cure; RAAS blockade (ACEi/ARB) started early reduces proteinuria and delays ESKD in Alport, and SGLT2 inhibitors are increasingly used adjuncts. Transplant is effective but rare anti-GBM (Goodpasture-like) post-transplant disease can occur.

BOARD TRAP

After kidney transplant, Alport males can develop de novo anti-GBM disease against the now-foreign normal α3(IV) collagen — a unique post-transplant complication.
